LIKITH PODALAKURU

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PROFILE SUMMARY

AI engineer with 3 years of experience building enterprise-scale C++ systems before transitioning to AI and machine learning. Experienced in developing end-to-end AI applications, computer vision solutions, and intelligent systems, with a focus on building reliable, production-ready software that delivers measurable results.

EDUCATION

Binghamton University, State University of New York, Thomas J. Watson College of Engineering
Master of Science in Computer Science (Artificial Intelligence Track) August 2024 - May 2026
Relevant Coursework: Machine Learning, Artificial Intelligence, System Programming, Deep Learning

Vellore Institute of Technology, Bangalore, India
Postgraduate Certification Program in Artificial Intelligence Aug 2023 – Jun 2024

SRM Valliammai Engineering College, Chennai, India
Bachelor of Engineering in Computer Science Aug 2016 – Apr 2020

TECHNICAL SKILLS

Languages: Python, C++, SQL, Rust
AI/ML: PyTorch, Scikit-learn, XGBoost, Reinforcement Learning, Computer Vision, RAG
Frameworks: FastAPI, Streamlit, OpenCV, FAISS, ONNX Runtime
Tools: Docker, Git, GitHub Actions, Linux, AWS
Databases: MySQL, MongoDB, Redis
Systems: Multithreading, Distributed Systems, NetworkX, OSMnx

PROFESSIONAL EXPERIENCE

Tata Consultancy Services - Systems Engineer (Client: Marriott International) Jan 2021 – Sep 2023

- Resolved critical production incidents by debugging C++ and Assembly systems supporting large-scale reservation platforms, restoring services under strict SLAs and minimizing downtime impact.
- Led daily incident management calls as the primary point of contact, coordinating with offshore teams and stakeholders to triage, prioritize, and resolve high-impact issues efficiently.
- Diagnosed and fixed a high-risk long-stay booking defect that could have resulted in an estimated $500K+ annual revenue loss, ensuring billing accuracy and system reliability.
- Performed unit and integration testing on code changes before formal handoff to the QA team, writing test cases and reproducing defects to validate fixes and prevent regressions in a policy-controlled release process.
- Automated repetitive operational tasks using scripting and analyzed logs to identify root causes, reducing recurring incidents and manual effort through structured documentation and reusable team tools.

PROJECT EXPERIENCE

Emergency Routing for Smart Response: Vision-Confirmed Routing on Live NYC Cameras June 2025 - Jul 2026

- Standard dispatch routing cannot see real-time blockages, so built a Manhattan emergency router that scores 371 live NYC DOT traffic cameras with YOLOv12 and converts congestion into edge weights on the real OpenStreetMap network (10,893 intersections, 24,852 street segments).
- Combined Dijkstra/A* planning with a vision gate that re-verifies every camera on the chosen route and replans mid-drive; cut mean simulated response time 20.8% (27.0 to 21.4 minutes) versus the static shortest-path baseline across 30 seeded emergency episodes; a corridor Q-learning agent trained under congestion noise scored 18.9%, reported honestly as slightly behind A*.
- Halved the deployed Docker image (3.27 GB to 1.5 GB) by replacing PyTorch inference with ONNX Runtime, removing a 728 MB dependency and reimplementing letterboxing and NMS by hand, verified with a parity test agreeing within 0.0012; shipped as a public Streamlit demo with offline fallbacks for every external dependency.

Multi-Agent Traffic Signal Control on Real Manhattan Streets (SUMO + MARL) July 2026

- Poor signal timing causes roughly 10% of US traffic delay, so built a pipeline converting raw OpenStreetMap extracts into validated SUMO scenarios (72 signalized Midtown intersections) with seeded, identical demand per controller; implemented fixed-time and max-pressure (Varaiya 2013) baselines, with max-pressure cutting mean time lost per vehicle 58% (131.6s to 55.3s).
- Trained one Double-DQN agent per intersection: learned agents cut time lost 45% versus fixed-time on held-out demand seeds, and neighbor-communicating agents beat independent agents on every held-out seed, closing about 40% of the remaining gap to max-pressure, directly quantifying the value of inter-agent communication.
- Enforced safety invariants (minimum green, yellow transitions) in the environment rather than the policy, with graceful degradation to fixed-time on any controller failure; next milestone scales the communication experiment to the full 72-intersection network.

Traffic Monitoring & Surveillance: Scalable Vehicle Detection December 2025

- Dense urban traffic makes manual monitoring impractical, so designed a deep-learning vehicle detection system and fine-tuned YOLOv11 on real-world traffic imagery, tuning augmentation and hyperparameters to hold detection quality across lighting, weather, and occlusion-heavy density variations.
- Delivered reliable detection across varying traffic conditions; this system became the perception foundation later extended into the live 371-camera NYC emergency routing project above.

Hashassin: Distributed Password Hashing & Cracking Tool (Rust) January 2025

- Built a multi-crate Rust CLI for password hashing and rainbow-table cracking (MD5, SHA-256, SHA-3, Scrypt) with a Tokio async client-server protocol, Rayon-parallelized compute, an LRU cache, and separate compute/async thread pools for tunable multi-core throughput.

Prediction of Malevolent Files: Machine Learning Threat Detection June 2024

- Signature-based tools miss novel binaries, so built a malware detection pipeline classifying executables as benign or malicious from large-scale static feature datasets using Scikit-learn.
- Engineered the full workflow (preprocessing, feature selection, model training and comparison) for high-dimensional, imbalanced feature spaces, improving detection through systematic evaluation rather than single-model tuning.

Impact of Electric Vehicles: Data Analysis and Predictive Modeling April 2024

- Analyzed real EV trip and charging data end to end: compared Linear, Ridge, and Gradient Boosting regressors on MSE and R^2, applied ARIMA forecasting with residual diagnostics for trip-distance trends, and used Isolation Forest to surface anomalous trips and data errors.
- Tuned models via grid search and translated results into insights on charging behavior and infrastructure planning; currently being upgraded into a production forecasting and smart-charging system on real Caltech ACN-Data.

ETA Prediction with Graph Neural Networks: 10M+ Real NYC Taxi Trips In Progress, Expected Aug 2026

- Next in the research arc: trip-time prediction over the real NYC road graph using GraphSAGE on 10M+ NYC TLC trip records, benchmarked against a tuned XGBoost baseline with per-borough error analysis, conformal prediction intervals, and a FastAPI endpoint with measured p95 latency.

PUBLICATIONS

Gottam, D., **Podalakuru, L.**, et al. *Reinforcement Learning for Autonomous Lunar Landing: A Comparative Analysis of Algorithm Performance*. iTech SECOM, 2025.

Ashwin R, **Podalakuru, L.**, Kumar, M.S. *"Proof of Work" Generation Using Blockchain*. International Journal of Advance Research and Innovative Ideas in Education, Vol. 6, 2020.